

PCS-302 : Data Structures Lab

Balanced Parentheses | Checking an Expression Using a Stack

Problem Statement

Given an expression string consisting of opening and closing brackets '{', '}', '(', ')', '[', ']', design an algorithm and a program to check whether this expression has balanced parentheses or not.

Example

Expression: { ([]) } --> BALANCED

Expression: { (]) } --> NOT BALANCED
(] does not match ()

Expression: (() --> NOT BALANCED
(one '(' left unmatched)

Algorithm (Using a Stack)

1. Create an empty stack.
2. For each character ch in the expression:
 - a. If ch is an opening bracket ((, { , [):
push(ch)
 - b. If ch is a closing bracket () , } ,]):
if stack is empty:
expression is NOT BALANCED, stop
pop the top element
if popped element does not match ch's opening bracket:
expression is NOT BALANCED, stop
3. After scanning all characters:
if stack is empty: expression is BALANCED
else: expression is NOT BALANCED

C Program

```
#include <stdio.h>
#include <string.h>
#define MAX 100

char stack[MAX];
int top = -1;

void push(char c) {
    top = top + 1;
    stack[top] = c;
}

char pop() {
    char c = stack[top];
    top = top - 1;
}
```

```

    return c;
}

int main() {
    char expr[MAX];
    int i;
    int balanced = 1;    // assume balanced initially

    printf("Enter expression: ");
    scanf("%s", expr);

    for (i = 0; expr[i] != '\0'; i++) {
        char ch = expr[i];

        if (ch == '(' || ch == '{' || ch == '[') {
            push(ch);
        }
        else if (ch == ')' || ch == '}' || ch == ']') {
            if (top == -1) {
                balanced = 0;    // closing bracket with nothing to match
                break;
            }
            char topCh = pop();
            if ((ch == ')' && topCh != '(') ||
                (ch == '}' && topCh != '{') ||
                (ch == ']' && topCh != '[')) {
                balanced = 0;    // mismatched pair
                break;
            }
        }
    }

    if (balanced == 1 && top == -1)
        printf("Expression is BALANCED.\n");
    else
        printf("Expression is NOT BALANCED.\n");

    return 0;
}

```